Kauaʻi stress-test annex

Data as of 2026-10-28

Planning aid. Supports, does not replace, county emergency plans. Not an official shelter list. Per-area counts from 1 to 9 are not shown. All figures are public Census Bureau, CDC and HHS estimates, not personal records.

Stress test, not a forecast. Each test changes one thing in the plan (a site closes, two hazards strike together, a road is cut, more people come) and counts who is affected, from the plan's own data. Deterministic code computes every number from the plan's data; people review the results.

People counted: residents of Census areas whose drive from the area's centre point to the nearest usable 2026 site was 15 minutes or less before the loss and is more than 15 minutes (or no route) after it; areas already beyond 15 minutes are not counted.

158 tests ran for Kauaʻi; 109 affect someone or need county data and are listed below. Two-site losses are listed only when they add people beyond either site alone. This is not a statement that the area has no shelter: counties designate and open shelters for each event, and many public buildings are not on the 2026 list.

Hurricane: top 10

1. **All four counties at once (hurricane)** (test 9: Several counties at once)
   43,518 people who may need a public shelter (likely), counties together (cots: 43,518; shelter staff per shift: 2,921).
   Cots: 43,518 · Counties: Kauai, Honolulu, Maui, Hawaii · People with limited English in shelter, by language (interpreters): Chinese: 247; Chuukese: 58; Hawaiian: 28; Ilocano: 369; Japanese: 201; Korean: 99; Marshallese: 24; Pidgin (Hawai‘i Creole English): 25; Spanish: 29; Tagalog/Filipino: 303; Vietnamese: 66 · Meals per day: 87,036 · People in shelter (likely): 43,518 · Rounding: counties' unrounded numbers added, then rounded up once · Shelter staff per shift: 2,921 · Water (gallons per day): 43,518
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: One rounding rule for counties together: add the unrounded county numbers, then round up once: round up.
2. **Lose Kapa‘a Middle School and Kaua‘i High School Gym (hurricane)** (test 2: Two sites lost)
   33,883 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 7; people more than losing either site alone: 10,099).
   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 7 · Drive limit (minutes): 15 · Island: Kaua‘i · People more than losing either site alone: 10,099 · People left with no usable 2026 site on the island: 0 · Sites: Kapa‘a Middle School, Kaua‘i High School Gym
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
3. **Lose Kalāheo Neighborhood Center and Kapa‘a Middle School (hurricane)** (test 2: Two sites lost)
   32,787 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 7; people more than losing either site alone: 9,003).
   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 7 · Drive limit (minutes): 15 · Island: Kaua‘i · People more than losing either site alone: 9,003 · People left with no usable 2026 site on the island: 0 · Sites: Kalāheo Neighborhood Center, Kapa‘a Middle School
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

4. **Kaua‘i and Honolulu at once (hurricane)** (test 9: Several counties at once)
   32,509 people who may need a public shelter (likely), counties together (cots: 32,509; shelter staff per shift: 2,177).
   Cots: 32,509 · Counties: Kauai, Honolulu · People with limited English in shelter, by language (interpreters): Chinese: 240; Chuukese: 58; Hawaiian: 17; Ilocano: 277; Japanese: 177; Korean: 99; Marshallese: 14; Spanish: 5; Tagalog/Filipino: 264; Vietnamese: 66 · Meals per day: 65,017 · People in shelter (likely): 32,509 · Rounding: counties' unrounded numbers added, then rounded up once · Shelter staff per shift: 2,177 · Water (gallons per day): 32,509
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: One rounding rule for counties together: add the unrounded county numbers, then round up once: round up.

5. **Stranded visitors for 3 or 7 days – Kaua‘i** (test 21: Stranded visitors)
   28,577 visitors on an average day (island level).
   By length of stay: 3 days: 171,462 meals, 85,731 gallons of water; 7 days: 400,078 meals, 200,039 gallons of water · Visitors on an average day (2025): 28,577
   Sources: DBEDT State of Hawai‘i Data Book 2025, Table 7.06 (average daily visitor census, 2025); Florida Division of Emergency Management, Sheltering Guidance Aid (2010). Team assumptions: Every visitor on the island is stranded and needs meals and water from mass care (upper bound): 100%.

6. **Lose Kapa‘a Middle School (hurricane)** (test 1: One site lost)
   23,784 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 5).
   Census areas pushed outside a 15-minute drive: 5 · Drive limit (minutes): 15 · Island: Kaua‘i · May need a public shelter (likely): 714 · Site: Kapa‘a Middle School
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

7. **Lose Kaua‘i High School Gym (hurricane)** (test 1: One site lost)
   10,099 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2).
   Census areas pushed outside a 15-minute drive: 2 · Drive limit (minutes): 15 · Island: Kaua‘i · May need a public shelter (likely): 303 · Site: Kaua‘i High School Gym
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

8. **Future sea-level-rise scenario (3.2 ft; State of Hawai‘i SLR-XA), tested against 2026 sites – Kaua‘i: every potentially flooded highway segment closed** (test 24: Road closures (driving))
   9,151 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+21.7 min to any known site; +0 min to a 2026 site).
   Census areas cut off: none · Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 0 · Census areas losing known sites within reach: 3 · Census areas with no route at all: 0 · Census areas routed: 4 · Census areas with a longer drive to a 2026 site: 0 · Cut off means: over 45 minutes or no route · Most minutes added to any known site: 21.7 · Most minutes added to the nearest 2026 site: 0 · Typical minutes added (median, where longer): 0 · People in areas cut off: 0 · People with a longer trip: 0 · People losing known sites within reach: 9,151 · Within reach means: 45-minute drive · Why this road:
   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS); Hawai‘i Sea Level Rise Viewer, potential flooded highways at 3.2 ft (PacIOOS, via Hawai‘i Statewide GIS). Team assumptions: Future sea-level-rise scenario (3.2 ft; State of Hawai‘i SLR-XA), tested against 2026 sites: every potentially flooded highway segment on the island closed at once: all segments.
9. **Kūhiō Highway (Route 560) near the Hanalei bridge** (test 24: Road closures (driving))
   7,793 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+0 min to any known site; +0 min to a 2026 site; cut off from every 2026 site: Hā‘ena / Hanalei · Census area 401.04).
   Census areas cut off: Hā‘ena / Hanalei · Census area 401.04 · Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 1 · Census areas losing known sites within reach: 3 · Census areas with no route at all: 0 · Census areas routed: 17 · Census areas with a longer drive to a 2026 site: 0 · Cut off means: over 45 minutes or no route · Most minutes added to any known site: 0 · Most minutes added to the nearest 2026 site: 0 · Typical minutes added (median, where longer): 0 · People in areas cut off: 1,278 · People with a longer trip: 0 · People losing known sites within reach: 7,793 · Within reach means: 45-minute drive · Why this road: Only road to Hanalei and the North Shore beyond it; one-lane bridge and flood-prone valley floor (team assumption).
   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS). Team assumptions: Closure location along the highway (a stretch of about 0.8 km): team assumption.

10. **Hurricane and inland flooding together – Kaua‘i** (test 5: Hurricane and inland flooding together)
    2,633 people who may need a public shelter (likely) (2026 sites usable for every hazard in the pair: 5; site gap: 6).

    Census areas: 17 · People if each hazard is counted separately: 94,915 · People in either footprint (counted once): 73,511 · May need a public shelter (likely): 2,633 · Site gap: 6 · Sites needed at 250 people per site (team assumption): 11 · 2026 sites usable for every hazard in the pair: 5

    Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: Sites needed at a given number of people per site (capacity unknown): 250; A person in two footprints is counted once, at the higher rate: highest rate.

Wildfire: top 10

1. **Stranded visitors for 3 or 7 days – Kauaʻi** (test 21: Stranded visitors)
   28,577 visitors on an average day (island level).
   By length of stay: 3 days: 171,462 meals, 85,731 gallons of water; 7 days: 400,078 meals, 200,039 gallons of water · Visitors on an average day (2025): 28,577
   Sources: DBEDT State of Hawaiʻi Data Book 2025, Table 7.06 (average daily visitor census, 2025); Florida Division of Emergency Management, Sheltering Guidance Aid (2010). Team assumptions: Every visitor on the island is stranded and needs meals and water from mass care (upper bound): 100%.
2. **All four counties at once (wildfire)** (test 9: Several counties at once)
   22,115 people who may need a public shelter (likely), counties together (cots: 22,115; shelter staff per shift: 1,534).
   Cots: 22,115 · Counties: Kauai, Honolulu, Maui, Hawaii · People with limited English in shelter, by language (interpreters): Chinese: 61; Chuukese: 14; Hawaiian: 9; Ilocano: 110; Japanese: 54; Korean: 24; Marshallese: 8; Pidgin (Hawaiʻi Creole English): 10; Spanish: 12; Tagalog/Filipino: 84; Vietnamese: 16 · Meals per day: 44,229 · People in shelter (likely): 22,115 · Rounding: counties' unrounded numbers added, then rounded up once · Shelter staff per shift: 1,534 · Water (gallons per day): 22,115
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: One rounding rule for counties together: add the unrounded county numbers, then round up once: round up.
3. **Kauaʻi and Honolulu at once (wildfire)** (test 9: Several counties at once)
   16,795 people who may need a public shelter (likely), counties together (cots: 16,795; shelter staff per shift: 1,163).
   Cots: 16,795 · Counties: Kauai, Honolulu · People with limited English in shelter, by language (interpreters): Chinese: 58; Chuukese: 14; Hawaiian: 4; Ilocano: 66; Japanese: 43; Korean: 24; Marshallese: 4; Spanish: 1; Tagalog/Filipino: 63; Vietnamese: 16 · Meals per day: 33,590 · People in shelter (likely): 16,795 · Rounding: counties' unrounded numbers added, then rounded up once · Shelter staff per shift: 1,163 · Water (gallons per day): 16,795
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: One rounding rule for counties together: add the unrounded county numbers, then round up once: round up.

4. **Areas high in the State Fire Risk layer but not in the plan's wildfire areas, and the reverse – Kauaʻi** (test 23b: State Fire Risk Areas vs the plan's wildfire areas)
   15,705 people in areas rated high only by the State Fire Risk layer (Census areas high in the State layer only: 3; Census areas in the plan's wildfire areas only: 4).
   Areas high in both, of all areas high in either: 12% · Census areas high in both: 1 · Census areas in the plan's wildfire areas only: 4 · Census areas high in the State layer only: 3 · Largest plan-only areas: Wailua Homesteads / Kapaʻa · Census area 402.04, Kīlauea · Census area 401.05, Lāwaʻi / ʻŌmaʻo · Census area 406.04, Kalāheo / Lāwaʻi · Census area 407.01 · People in plan-only areas: 17,148 · People in State-only areas: 15,705 · People in the plan's wildfire areas (FEMA National Risk Index): 21,633 · People in State High fire risk areas: 21,359 · Largest State-only areas: Līhuʻe · Census area 405, Kapaʻa · Census area 403.01, Kapaʻa · Census area 403.02
   Sources: State Fire Risk Areas (DLNR Division of Forestry and Wildlife, via Hawaiʻi Statewide GIS); FEMA National Risk Index wildfire scores; CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks. Team assumptions: An area is 'high' in the State layer when at least half its people live in a High fire risk area: 50%.
5. **Lose Kapaʻa Middle School and Kīlauea Neighborhood Center (wildfire)** (test 2: Two sites lost)
   15,032 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 3; people more than losing either site alone: 4,508).
   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 3 · Drive limit (minutes): 15 · Island: Kauaʻi · People more than losing either site alone: 4,508 · People left with no usable 2026 site on the island: 0 · Sites: Kapaʻa Middle School, Kīlauea Neighborhood Center
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawaiʻi Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
6. **Lose Kalāheo Neighborhood Center and Kapaʻa Middle School (wildfire)** (test 2: Two sites lost)
   14,033 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 3; people more than losing either site alone: 3,509).
   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 3 · Drive limit (minutes): 15 · Island: Kauaʻi · People more than losing either site alone: 3,509 · People left with no usable 2026 site on the island: 0 · Sites: Kalāheo Neighborhood Center, Kapaʻa Middle School
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawaiʻi Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
7. **Lose Kapaʻa Middle School (wildfire)** (test 1: One site lost)
   10,524 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2).
   Census areas pushed outside a 15-minute drive: 2 · Drive limit (minutes): 15 · Island: Kauaʻi · May need a public shelter (likely): 526 · Site: Kapaʻa Middle School
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawaiʻi Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

8. **Future sea-level-rise scenario (3.2 ft; State of Hawai‘i SLR-XA), tested against 2026 sites – Kaua‘i: every potentially flooded highway segment closed** (test 24: Road closures (driving))
   9,151 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+21.7 min to any known site; +0 min to a 2026 site).
   Census areas cut off: none · Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 0 · Census areas losing known sites within reach: 3 · Census areas with no route at all: 0 · Census areas routed: 4 · Census areas with a longer drive to a 2026 site: 0 · Cut off means: over 45 minutes or no route · Most minutes added to any known site: 21.7 · Most minutes added to the nearest 2026 site: 0 · Typical minutes added (median, where longer): 0 · People in areas cut off: 0 · People with a longer trip: 0 · People losing known sites within reach: 9,151 · Within reach means: 45-minute drive · Why this road:
   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS); Hawai‘i Sea Level Rise Viewer, potential flooded highways at 3.2 ft (PacIOOS, via Hawai‘i Statewide GIS). Team assumptions: Future sea-level-rise scenario (3.2 ft; State of Hawai‘i SLR-XA), tested against 2026 sites: every potentially flooded highway segment on the island closed at once: all segments.
9. **Kūhiō Highway (Route 560) near the Hanalei bridge** (test 24: Road closures (driving))
   7,793 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+0 min to any known site; +0 min to a 2026 site; cut off from every 2026 site: Hā‘ena / Hanalei · Census area 401.04).
   Census areas cut off: Hā‘ena / Hanalei · Census area 401.04 · Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 1 · Census areas losing known sites within reach: 3 · Census areas with no route at all: 0 · Census areas routed: 17 · Census areas with a longer drive to a 2026 site: 0 · Cut off means: over 45 minutes or no route · Most minutes added to any known site: 0 · Most minutes added to the nearest 2026 site: 0 · Typical minutes added (median, where longer): 0 · People in areas cut off: 1,278 · People with a longer trip: 0 · People losing known sites within reach: 7,793 · Within reach means: 45-minute drive · Why this road: Only road to Hanalei and the North Shore beyond it; one-lane bridge and flood-prone valley floor (team assumption).
   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS). Team assumptions: Closure location along the highway (a stretch of about 0.8 km): team assumption.
10. **Lose Kīlauea Neighborhood Center (wildfire)** (test 1: One site lost)
    4,508 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1).
    Census areas pushed outside a 15-minute drive: 1 · Drive limit (minutes): 15 · Island: Kaua‘i · May need a public shelter (likely): 225 · Site: Kīlauea Neighborhood Center
    Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

Tsunami: top 10

1. **Visitors on the island(s) – Kauaʻi (tsunami)** (test 20: Visitors in tsunami zones)
   28,577 visitors on an average day (island level) (visitors inside the tsunami zone: needs county data (hotel and rental locations)).
   Detail: Kauaʻi · Peak day: needs DBEDT monthly data (the Data Book gives the annual average) · Visitors on an average day (2025): 28,577 · Visitors inside the tsunami zone: needs county data (hotel and rental locations)
   Sources: DBEDT State of Hawaiʻi Data Book 2025, Table 7.06 (average daily visitor census, 2025).
2. **Stranded visitors for 3 or 7 days – Kauaʻi** (test 21: Stranded visitors)
   28,577 visitors on an average day (island level).
   By length of stay: 3 days: 171,462 meals, 85,731 gallons of water; 7 days: 400,078 meals, 200,039 gallons of water · Visitors on an average day (2025): 28,577
   Sources: DBEDT State of Hawaiʻi Data Book 2025, Table 7.06 (average daily visitor census, 2025); Florida Division of Emergency Management, Sheltering Guidance Aid (2010). Team assumptions: Every visitor on the island is stranded and needs meals and water from mass care (upper bound): 100%.
3. **Extreme tsunami zone (worst case) – Kauaʻi** (test 8: Extreme tsunami zone (worst case))
   9,862 people in the extreme tsunami evacuation zone (may need a public shelter (likely): 493; 2026 sites inside the extreme zone: 1).
   People in the extreme tsunami zone: 9,862 · People in the standard tsunami zone: 9,034 · May need a shelter (high): 1,085 · May need a public shelter (likely): 493 · May need a shelter (low): 197 · 2026 sites inside the extreme zone: 1 · Sites needed at 250 people per site (team assumption): 2 · 2026 sites usable outside both zones: 4
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; State tsunami evacuation zones and extreme tsunami evacuation zones (Hawaiʻi Statewide GIS); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawaiʻi Statewide GIS); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: Sites needed at a given number of people per site (capacity unknown): 250.

4. **Future sea-level-rise scenario (3.2 ft; State of Hawai‘i SLR-XA), tested against 2026 sites – Kaua‘i: every potentially flooded highway segment closed** (test 24: Road closures (driving))
   9,151 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+21.7 min to any known site; +0 min to a 2026 site).

   Census areas cut off: none · Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 0 · Census areas losing known sites within reach: 3 · Census areas with no route at all: 0 · Census areas routed: 4 · Census areas with a longer drive to a 2026 site: 0 · Cut off means: over 45 minutes or no route · Most minutes added to any known site: 21.7 · Most minutes added to the nearest 2026 site: 0 · Typical minutes added (median, where longer): 0 · People in areas cut off: 0 · People with a longer trip: 0 · People losing known sites within reach: 9,151 · Within reach means: 45-minute drive · Why this road:

   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS); Hawai‘i Sea Level Rise Viewer, potential flooded highways at 3.2 ft (PacIOOS, via Hawai‘i Statewide GIS). Team assumptions: Future sea-level-rise scenario (3.2 ft; State of Hawai‘i SLR-XA), tested against 2026 sites: every potentially flooded highway segment on the island closed at once: all segments.

5. **Kūhiō Highway (Route 560) near the Hanalei bridge** (test 24: Road closures (driving))
   7,793 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+0 min to any known site; +0 min to a 2026 site; cut off from every 2026 site: Hā‘ena / Hanalei · Census area 401.04).

   Census areas cut off: Hā‘ena / Hanalei · Census area 401.04 · Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 1 · Census areas losing known sites within reach: 3 · Census areas with no route at all: 0 · Census areas routed: 17 · Census areas with a longer drive to a 2026 site: 0 · Cut off means: over 45 minutes or no route · Most minutes added to any known site: 0 · Most minutes added to the nearest 2026 site: 0 · Typical minutes added (median, where longer): 0 · People in areas cut off: 1,278 · People with a longer trip: 0 · People losing known sites within reach: 7,793 · Within reach means: 45-minute drive · Why this road: Only road to Hanalei and the North Shore beyond it; one-lane bridge and flood-prone valley floor (team assumption).

   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS). Team assumptions: Closure location along the highway (a stretch of about 0.8 km): team assumption.

6. **All four counties at once (tsunami)** (test 9: Several counties at once)
   6,382 people who may need a public shelter (likely), counties together (cots: 6,382; shelter staff per shift: 442).
   Cots: 6,382 · Counties: Kauai, Honolulu, Maui, Hawaii · People with limited English in shelter, by language (interpreters): Chinese: 31; Chuukese: 8; Hawaiian: 5; Ilocano: 60; Japanese: 28; Korean: 13; Marshallese: 4; Pidgin (Hawai‘i Creole English): 4; Spanish: 7; Tagalog/Filipino: 46; Vietnamese: 9 · Meals per day: 12,764 · People in shelter (likely): 6,382 · Rounding: counties' unrounded numbers added, then rounded up once · Shelter staff per shift: 442 · Water (gallons per day): 6,382
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: One rounding rule for counties together: add the unrounded county numbers, then round up once: round up.
7. **Kaua‘i and Honolulu at once (tsunami)** (test 9: Several counties at once)
   4,469 people who may need a public shelter (likely), counties together (cots: 4,469; shelter staff per shift: 306).
   Cots: 4,469 · Counties: Kauai, Honolulu · People with limited English in shelter, by language (interpreters): Chinese: 30; Chuukese: 8; Hawaiian: 3; Ilocano: 37; Japanese: 23; Korean: 13; Marshallese: 2; Spanish: 1; Tagalog/Filipino: 34; Vietnamese: 9 · Meals per day: 8,938 · People in shelter (likely): 4,469 · Rounding: counties' unrounded numbers added, then rounded up once · Shelter staff per shift: 306 · Water (gallons per day): 4,469
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: One rounding rule for counties together: add the unrounded county numbers, then round up once: round up.
8. **Lose Kapa‘a Middle School and Kaua‘i High School Gym (tsunami)** (test 2: Two sites lost)
   2,951 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 5; people more than losing either site alone: 138).
   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 5 · Drive limit (minutes): 15 · Island: Kaua‘i · People more than losing either site alone: 138 · People left with no usable 2026 site on the island: 0 · Sites: Kapa‘a Middle School, Kaua‘i High School Gym
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

9. **Lose Kalāheo Neighborhood Center and Kapa‘a Middle School (tsunami)** (test 2: Two sites lost)

   2,901 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 4; people more than losing either site alone: 88).

   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 4 · Drive limit (minutes): 15 · Island: Kaua‘i · People more than losing either site alone: 88 · People left with no usable 2026 site on the island: 0 · Sites: Kalāheo Neighborhood Center, Kapa‘a Middle School

   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

10. **Lose Kapa‘a Middle School (tsunami)** (test 1: One site lost)

    2,813 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 3).

    Census areas pushed outside a 15-minute drive: 3 · Drive limit (minutes): 15 · Island: Kaua‘i · May need a public shelter (likely): 141 · Site: Kapa‘a Middle School

    Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

Inland flooding: top 10

1. **Stranded visitors for 3 or 7 days – Kauaʻi** (test 21: Stranded visitors)
   28,577 visitors on an average day (island level).
   By length of stay: 3 days: 171,462 meals, 85,731 gallons of water; 7 days: 400,078 meals, 200,039 gallons of water · Visitors on an average day (2025): 28,577
   Sources: DBEDT State of Hawaiʻi Data Book 2025, Table 7.06 (average daily visitor census, 2025); Florida Division of Emergency Management, Sheltering Guidance Aid (2010). Team assumptions: Every visitor on the island is stranded and needs meals and water from mass care (upper bound): 100%.
2. **All four counties at once (inland flooding)** (test 9: Several counties at once)
   25,452 people who may need a public shelter (likely), counties together (cots: 25,452; shelter staff per shift: 1,723).
   Cots: 25,452 · Counties: Kauai, Honolulu, Maui, Hawaii · People with limited English in shelter, by language (interpreters): Chinese: 100; Chuukese: 24; Hawaiian: 12; Ilocano: 153; Japanese: 82; Korean: 40; Marshallese: 10; Pidgin (Hawaiʻi Creole English): 12; Spanish: 13; Tagalog/Filipino: 124; Vietnamese: 27 · Meals per day: 50,904 · People in shelter (likely): 25,452 · Rounding: counties' unrounded numbers added, then rounded up once · Shelter staff per shift: 1,723 · Water (gallons per day): 25,452
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: One rounding rule for counties together: add the unrounded county numbers, then round up once: round up.
3. **Kauaʻi and Honolulu at once (inland flooding)** (test 9: Several counties at once)
   19,544 people who may need a public shelter (likely), counties together (cots: 19,544; shelter staff per shift: 1,322).
   Cots: 19,544 · Counties: Kauai, Honolulu · People with limited English in shelter, by language (interpreters): Chinese: 97; Chuukese: 24; Hawaiian: 7; Ilocano: 112; Japanese: 72; Korean: 40; Marshallese: 6; Spanish: 2; Tagalog/Filipino: 107; Vietnamese: 27 · Meals per day: 39,087 · People in shelter (likely): 19,544 · Rounding: counties' unrounded numbers added, then rounded up once · Shelter staff per shift: 1,322 · Water (gallons per day): 19,544
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: One rounding rule for counties together: add the unrounded county numbers, then round up once: round up.

4. **Lose Kapa‘a Middle School and Kaua‘i High School Gym (inland flooding)** (test 2: Two sites lost)

   11,982 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2; people more than losing either site alone: 4,681).

   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 2 · Drive limit (minutes): 15 · Island: Kaua‘i · People more than losing either site alone: 4,681 · People left with no usable 2026 site on the island: 0 · Sites: Kapa‘a Middle School, Kaua‘i High School Gym

   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

5. **Future sea-level-rise scenario (3.2 ft; State of Hawai‘i SLR-XA), tested against 2026 sites – Kaua‘i: every potentially flooded highway segment closed** (test 24: Road closures (driving))

   9,151 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+21.7 min to any known site; +0 min to a 2026 site).

   Census areas cut off: none · Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 0 · Census areas losing known sites within reach: 3 · Census areas with no route at all: 0 · Census areas routed: 4 · Census areas with a longer drive to a 2026 site: 0 · Cut off means: over 45 minutes or no route · Most minutes added to any known site: 21.7 · Most minutes added to the nearest 2026 site: 0 · Typical minutes added (median, where longer): 0 · People in areas cut off: 0 · People with a longer trip: 0 · People losing known sites within reach: 9,151 · Within reach means: 45-minute drive · Why this road:

   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS); Hawai‘i Sea Level Rise Viewer, potential flooded highways at 3.2 ft (PacIOOS, via Hawai‘i Statewide GIS). Team assumptions: Future sea-level-rise scenario (3.2 ft; State of Hawai‘i SLR-XA), tested against 2026 sites: every potentially flooded highway segment on the island closed at once: all segments.

6. **Kūhiō Highway (Route 560) near the Hanalei bridge** (test 24: Road closures (driving))
   7,793 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+0 min to any known site; +0 min to a 2026 site; cut off from every 2026 site: Hā‘ena / Hanalei · Census area 401.04).
   Census areas cut off: Hā‘ena / Hanalei · Census area 401.04 · Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 1 · Census areas losing known sites within reach: 3 · Census areas with no route at all: 0 · Census areas routed: 17 · Census areas with a longer drive to a 2026 site: 0 · Cut off means: over 45 minutes or no route · Most minutes added to any known site: 0 · Most minutes added to the nearest 2026 site: 0 · Typical minutes added (median, where longer): 0 · People in areas cut off: 1,278 · People with a longer trip: 0 · People losing known sites within reach: 7,793 · Within reach means: 45-minute drive · Why this road: Only road to Hanalei and the North Shore beyond it; one-lane bridge and flood-prone valley floor (team assumption).
   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS). Team assumptions: Closure location along the highway (a stretch of about 0.8 km): team assumption.
7. **Lose Kaua‘i High School Gym (inland flooding)** (test 1: One site lost)
   7,301 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1).
   Census areas pushed outside a 15-minute drive: 1 · Drive limit (minutes): 15 · Island: Kaua‘i · May need a public shelter (likely): 365 · Site: Kaua‘i High School Gym
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
8. **Lose Kalāheo Neighborhood Center and Kaua‘i High School Gym (inland flooding)** (test 2: Two sites lost)
   7,301 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1; people more than losing either site alone: 0).
   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 1 · Drive limit (minutes): 15 · Island: Kaua‘i · People more than losing either site alone: 0 · People left with no usable 2026 site on the island: 0 · Sites: Kalāheo Neighborhood Center, Kaua‘i High School Gym
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

9. **Lose Kapa‘a Middle School (inland flooding)** (test 1: One site lost)
   4,681 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1).
   Census areas pushed outside a 15-minute drive: 1 · Drive limit (minutes): 15 · Island: Kaua‘i · May need a public shelter (likely): 234 · Site: Kapa‘a Middle School
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
10. **Hurricane and inland flooding together – Kaua‘i** (test 5: Hurricane and inland flooding together)
    2,633 people who may need a public shelter (likely) (2026 sites usable for every hazard in the pair: 5; site gap: 6).
    Census areas: 17 · People if each hazard is counted separately: 94,915 · People in either footprint (counted once): 73,511 · May need a public shelter (likely): 2,633 · Site gap: 6 · Sites needed at 250 people per site (team assumption): 11 · 2026 sites usable for every hazard in the pair: 5
    Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: Sites needed at a given number of people per site (capacity unknown): 250; A person in two footprints is counted once, at the higher rate: highest rate.

Coastal flooding: top 10

1. **Stranded visitors for 3 or 7 days – Kauaʻi** (test 21: Stranded visitors)
   28,577 visitors on an average day (island level).
   By length of stay: 3 days: 171,462 meals, 85,731 gallons of water; 7 days: 400,078 meals, 200,039 gallons of water · Visitors on an average day (2025): 28,577
   Sources: DBEDT State of Hawaiʻi Data Book 2025, Table 7.06 (average daily visitor census, 2025); Florida Division of Emergency Management, Sheltering Guidance Aid (2010). Team assumptions: Every visitor on the island is stranded and needs meals and water from mass care (upper bound): 100%.
2. **All four counties at once (coastal flooding)** (test 9: Several counties at once)
   17,662 people who may need a public shelter (likely), counties together (cots: 17,662; shelter staff per shift: 1,171).
   Cots: 17,662 · Counties: Kauai, Honolulu, Maui, Hawaii · People with limited English in shelter, by language (interpreters): Chinese: 99; Chuukese: 23; Hawaiian: 13; Ilocano: 154; Japanese: 83; Korean: 39; Marshallese: 11; Pidgin (Hawaiʻi Creole English): 15; Spanish: 15; Tagalog/Filipino: 122; Vietnamese: 26 · Meals per day: 35,324 · People in shelter (likely): 17,662 · Rounding: counties' unrounded numbers added, then rounded up once · Shelter staff per shift: 1,171 · Water (gallons per day): 17,662
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: One rounding rule for counties together: add the unrounded county numbers, then round up once: round up.
3. **Kauaʻi and Honolulu at once (coastal flooding)** (test 9: Several counties at once)
   13,130 people who may need a public shelter (likely), counties together (cots: 13,130; shelter staff per shift: 868).
   Cots: 13,130 · Counties: Kauai, Honolulu · People with limited English in shelter, by language (interpreters): Chinese: 94; Chuukese: 23; Hawaiian: 7; Ilocano: 109; Japanese: 70; Korean: 39; Marshallese: 6; Spanish: 2; Tagalog/Filipino: 104; Vietnamese: 26 · Meals per day: 26,260 · People in shelter (likely): 13,130 · Rounding: counties' unrounded numbers added, then rounded up once · Shelter staff per shift: 868 · Water (gallons per day): 13,130
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: One rounding rule for counties together: add the unrounded county numbers, then round up once: round up.

4. **Lose Kapa‘a Middle School and Kaua‘i High School Gym (coastal flooding)** (test 2: Two sites lost)

   9,764 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2; people more than losing either site alone: 2,463).

   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 2 · Drive limit (minutes): 15 · Island: Kaua‘i · People more than losing either site alone: 2,463 · People left with no usable 2026 site on the island: 0 · Sites: Kapa‘a Middle School, Kaua‘i High School Gym

   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

5. **Future sea-level-rise scenario (3.2 ft; State of Hawai‘i SLR-XA), tested against 2026 sites – Kaua‘i: every potentially flooded highway segment closed** (test 24: Road closures (driving))

   9,151 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+21.7 min to any known site; +0 min to a 2026 site).

   Census areas cut off: none · Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 0 · Census areas losing known sites within reach: 3 · Census areas with no route at all: 0 · Census areas routed: 4 · Census areas with a longer drive to a 2026 site: 0 · Cut off means: over 45 minutes or no route · Most minutes added to any known site: 21.7 · Most minutes added to the nearest 2026 site: 0 · Typical minutes added (median, where longer): 0 · People in areas cut off: 0 · People with a longer trip: 0 · People losing known sites within reach: 9,151 · Within reach means: 45-minute drive · Why this road:

   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS); Hawai‘i Sea Level Rise Viewer, potential flooded highways at 3.2 ft (PacIOOS, via Hawai‘i Statewide GIS). Team assumptions: Future sea-level-rise scenario (3.2 ft; State of Hawai‘i SLR-XA), tested against 2026 sites: every potentially flooded highway segment on the island closed at once: all segments.

6. **Kūhiō Highway (Route 560) near the Hanalei bridge** (test 24: Road closures (driving))
   7,793 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+0 min to any known site; +0 min to a 2026 site; cut off from every 2026 site: Hā‘ena / Hanalei · Census area 401.04).
   Census areas cut off: Hā‘ena / Hanalei · Census area 401.04 · Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 1 · Census areas losing known sites within reach: 3 · Census areas with no route at all: 0 · Census areas routed: 17 · Census areas with a longer drive to a 2026 site: 0 · Cut off means: over 45 minutes or no route · Most minutes added to any known site: 0 · Most minutes added to the nearest 2026 site: 0 · Typical minutes added (median, where longer): 0 · People in areas cut off: 1,278 · People with a longer trip: 0 · People losing known sites within reach: 7,793 · Within reach means: 45-minute drive · Why this road: Only road to Hanalei and the North Shore beyond it; one-lane bridge and flood-prone valley floor (team assumption).
   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS). Team assumptions: Closure location along the highway (a stretch of about 0.8 km): team assumption.
7. **Lose Kaua‘i High School Gym (coastal flooding)** (test 1: One site lost)
   7,301 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1).
   Census areas pushed outside a 15-minute drive: 1 · Drive limit (minutes): 15 · Island: Kaua‘i · May need a public shelter (likely): 365 · Site: Kaua‘i High School Gym
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
8. **Lose Kalāheo Neighborhood Center and Kaua‘i High School Gym (coastal flooding)** (test 2: Two sites lost)
   7,301 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1; people more than losing either site alone: 0).
   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 1 · Drive limit (minutes): 15 · Island: Kaua‘i · People more than losing either site alone: 0 · People left with no usable 2026 site on the island: 0 · Sites: Kalāheo Neighborhood Center, Kaua‘i High School Gym
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

9. **Lose Kapa‘a Middle School (coastal flooding)** (test 1: One site lost)

   2,463 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1).

   Census areas pushed outside a 15-minute drive: 1 · Drive limit (minutes): 15 · Island: Kaua‘i · May need a public shelter (likely): 123 · Site: Kapa‘a Middle School

   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

10. **Rate at which demand needs more sites than are usable – Kaua‘i (coastal flooding)** (test 10: Shelter-seeking rate higher than planned)

    1,343 people who may need a shelter at the rate where usable sites run out (rate when usable sites run out: 7%; usable 2026 sites: 5).

    Known 2026 sites in the county: 5 · Not reached by: – · People in the risk area: 19,186 · Likely rate in the plan: 5% · Rate when usable sites run out: 7% · Sites needed by rate (250 people per site): 10%: 8; 11%: 9; 12%: 10; 13%: 10; 14%: 11; 15%: 12; 16%: 13; 17%: 14; 18%: 14; 19%: 15; 2%: 2; 20%: 16; 3%: 3; 4%: 4; 5%: 4; 6%: 5; 7%: 6; 8%: 7; 9%: 7 · Usable 2026 sites: 5

    Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: Sites needed at a given number of people per site (capacity unknown): 250; Rates tested: 2–20% in 1-point steps.

Every result, ranked by people affected

Rank	Test	Scenario	Hazard	Result
1	9	All four counties at once (hurricane)	Hurricane	43,518 people who may need a public shelter (likely), counties together (cots: 43,518; shelter staff per shift: 2,921)
1	2	Lose Kapaʻa Middle School and Kauaʻi High School Gym (hurricane)	Hurricane	33,883 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 7; people more than losing either site alone: 10,099)
2	2	Lose Kalāheo Neighborhood Center and Kapaʻa Middle School (hurricane)	Hurricane	32,787 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 7; people more than losing either site alone: 9,003)
3	2	Lose Kapaʻa Middle School and Kīlauea Neighborhood Center (hurricane)	Hurricane	30,299 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 7; people more than losing either site alone: 6,515)
4	9	Kauaʻi and Honolulu at once (hurricane)	Hurricane	32,509 people who may need a public shelter (likely), counties together (cots: 32,509; shelter staff per shift: 2,177)
4	20	Visitors on the island(s) – Kauaʻi (tsunami)	Tsunami	28,577 visitors on an average day (island level) (visitors inside the tsunami zone: needs county data (hotel and rental locations))
5	21	Stranded visitors for 3 or 7 days – Kauaʻi	All hazards	28,577 visitors on an average day (island level)
5	9	All four counties at once (inland flooding)	Inland flooding	25,452 people who may need a public shelter (likely), counties together (cots: 25,452; shelter staff per shift: 1,723)

Rank	Test	Scenario	Hazard	Result
6	1	Lose Kapaʻa Middle School (hurricane)	Hurricane	23,784 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 5)
6	9	All four counties at once (wildfire)	Wildfire	22,115 people who may need a public shelter (likely), counties together (cots: 22,115; shelter staff per shift: 1,534)
8	2	Lose Kalāheo Neighborhood Center and Kauaʻi High School Gym (hurricane)	Hurricane	19,102 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 4; people more than losing either site alone: 9,003)
9	9	Kauaʻi and Honolulu at once (inland flooding)	Inland flooding	19,544 people who may need a public shelter (likely), counties together (cots: 19,544; shelter staff per shift: 1,322)
9	2	Lose Kauaʻi High School Gym and Kīlauea Neighborhood Center (hurricane)	Hurricane	16,614 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 4; people more than losing either site alone: 6,515)
10	23b	Areas high in the State Fire Risk layer but not in the plan's wildfire areas, and the reverse – Kauaʻi	Wildfire	15,705 people in areas rated high only by the State Fire Risk layer (Census areas high in the State layer only: 3; Census areas in the plan's wildfire areas only: 4)
11	2	Lose Kalāheo Neighborhood Center and Kīlauea Neighborhood Center (hurricane)	Hurricane	15,518 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 4; people more than losing either site alone: 6,515)
12	9	All four counties at once (coastal flooding)	Coastal flooding	17,662 people who may need a public shelter (likely), counties together (cots: 17,662; shelter staff per shift: 1,171)

Rank	Test	Scenario	Hazard	Result
12	2	Lose Kapaʻa Middle School and Kīlauea Neighborhood Center (wildfire)	Wildfire	15,032 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 3; people more than losing either site alone: 4,508)
13	9	Kauaʻi and Honolulu at once (wildfire)	Wildfire	16,795 people who may need a public shelter (likely), counties together (cots: 16,795; shelter staff per shift: 1,163)
13	2	Lose Kalāheo Neighborhood Center and Kapaʻa Middle School (wildfire)	Wildfire	14,033 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 3; people more than losing either site alone: 3,509)
14	2	Lose Kapaʻa Middle School and Kauaʻi High School Gym (inland flooding)	Inland flooding	11,982 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2; people more than losing either site alone: 4,681)
15	1	Lose Kapaʻa Middle School (wildfire)	Wildfire	10,524 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2)
16	9	Kauaʻi and Honolulu at once (coastal flooding)	Coastal flooding	13,130 people who may need a public shelter (likely), counties together (cots: 13,130; shelter staff per shift: 868)
18	1	Lose Kauaʻi High School Gym (hurricane)	Hurricane	10,099 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2)
18	9	Kauaʻi and Hawaiʻi at once (hurricane)	Hurricane	8,271 people who may need a public shelter (likely), counties together (cots: 8,271; shelter staff per shift: 564)

Rank	Test	Scenario	Hazard	Result
19	9	Kauaʻi and Maui at once (hurricane)	Hurricane	7,150 people who may need a public shelter (likely), counties together (cots: 7,150; shelter staff per shift: 488)
20	8	Extreme tsunami zone (worst case) – Kauaʻi	Tsunami	9,862 people in the extreme tsunami evacuation zone (may need a public shelter (likely): 493; 2026 sites inside the extreme zone: 1)
20	9	All four counties at once (tsunami)	Tsunami	6,382 people who may need a public shelter (likely), counties together (cots: 6,382; shelter staff per shift: 442)
21	2	Lose Kapaʻa Middle School and Kauaʻi High School Gym (coastal flooding)	Coastal flooding	9,764 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2; people more than losing either site alone: 2,463)
22	24	Future sea-level-rise scenario (3.2 ft; State of Hawaiʻi SLR-XA), tested against 2026 sites – Kauaʻi: every potentially flooded highway segment closed	All hazards	9,151 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+21.7 min to any known site; +0 min to a 2026 site)
23	1	Lose Kalāheo Neighborhood Center (hurricane)	Hurricane	9,003 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2)
25	2	Lose Kalāheo Neighborhood Center and Kīlauea Neighborhood Center (wildfire)	Wildfire	8,017 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2; people more than losing either site alone: 3,509)
26	24	Kūhiō Highway (Route 560) near the Hanalei bridge	All hazards	7,793 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+0 min to any known site; +0 min to a 2026 site; cut off from every 2026 site: Hāʻena / Hanalei · Census area 401.04)

Rank	Test	Scenario	Hazard	Result
26	9	Kauaʻi and Honolulu at once (tsunami)	Tsunami	4,469 people who may need a public shelter (likely), counties together (cots: 4,469; shelter staff per shift: 306)
27	1	Lose Kauaʻi High School Gym (coastal flooding)	Coastal flooding	7,301 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1)
27	9	Kauaʻi and Hawaiʻi at once (inland flooding)	Inland flooding	4,347 people who may need a public shelter (likely), counties together (cots: 4,347; shelter staff per shift: 300)
28	1	Lose Kauaʻi High School Gym (inland flooding)	Inland flooding	7,301 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1)
28	9	Kauaʻi and Hawaiʻi at once (wildfire)	Wildfire	3,960 people who may need a public shelter (likely), counties together (cots: 3,960; shelter staff per shift: 276)
29	9	Kauaʻi and Maui at once (inland flooding)	Inland flooding	3,703 people who may need a public shelter (likely), counties together (cots: 3,703; shelter staff per shift: 255)
30	9	Kauaʻi and Hawaiʻi at once (coastal flooding)	Coastal flooding	3,658 people who may need a public shelter (likely), counties together (cots: 3,658; shelter staff per shift: 245)
31	9	Kauaʻi and Maui at once (wildfire)	Wildfire	3,524 people who may need a public shelter (likely), counties together (cots: 3,524; shelter staff per shift: 249)
32	9	Kauaʻi and Maui at once (coastal flooding)	Coastal flooding	2,794 people who may need a public shelter (likely), counties together (cots: 2,794; shelter staff per shift: 194)
34	9	Kauaʻi and Maui at once (tsunami)	Tsunami	1,760 people who may need a public shelter (likely), counties together (cots: 1,760; shelter staff per shift: 126)

Rank	Test	Scenario	Hazard	Result
35	1	Lose Kīlauea Neighborhood Center (hurricane)	Hurricane	6,515 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2)
35	9	Kauaʻi and Hawaiʻi at once (tsunami)	Tsunami	1,057 people who may need a public shelter (likely), counties together (cots: 1,057; shelter staff per shift: 80)
37	1	Lose Kapaʻa Middle School (inland flooding)	Inland flooding	4,681 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1)
41	1	Lose Kīlauea Neighborhood Center (wildfire)	Wildfire	4,508 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1)
44	1	Lose Kalāheo Neighborhood Center (wildfire)	Wildfire	3,509 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1)
47	2	Lose Kapaʻa Middle School and Kauaʻi High School Gym (tsunami)	Tsunami	2,951 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 5; people more than losing either site alone: 138)
48	2	Lose Kalāheo Neighborhood Center and Kapaʻa Middle School (tsunami)	Tsunami	2,901 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 4; people more than losing either site alone: 88)
49	2	Lose Kapaʻa Middle School and Kīlauea Neighborhood Center (tsunami)	Tsunami	2,858 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 4; people more than losing either site alone: 45)

Rank	Test	Scenario	Hazard	Result
50	1	Lose Kapaʻa Middle School (tsunami)	Tsunami	2,813 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 3)
52	5	Hurricane and inland flooding together – Kauaʻi	Hurricane + Inland flooding	2,633 people who may need a public shelter (likely) (2026 sites usable for every hazard in the pair: 5; site gap: 6)
53	1	Lose Kapaʻa Middle School (coastal flooding)	Coastal flooding	2,463 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1)
57	11	Stays of 3 to 30 days – Kauaʻi (hurricane)	Hurricane	2,205 people who may need a public shelter (likely)
58	10	Rate at which demand needs more sites than are usable – Kauaʻi (hurricane)	Hurricane	1,470 people who may need a shelter at the rate where usable sites run out (rate when usable sites run out: 2%; usable 2026 sites: 5)
59	10	Rate at which demand needs more sites than are usable – Kauaʻi (coastal flooding)	Coastal flooding	1,343 people who may need a shelter at the rate where usable sites run out (rate when usable sites run out: 7%; usable 2026 sites: 5)
60	7	Dam failure and inland flooding together – Kauaʻi	Dam failure + Inland flooding	1,317 people who may need a public shelter (likely) (2026 sites usable for every hazard in the pair: 5; site gap: 1)
61	10	Rate at which demand needs more sites than are usable – Kauaʻi (wildfire)	Wildfire	1,298 people who may need a shelter at the rate where usable sites run out (rate when usable sites run out: 6%; usable 2026 sites: 5)
62	10	Rate at which demand needs more sites than are usable – Kauaʻi (inland flooding)	Inland flooding	1,284 people who may need a shelter at the rate where usable sites run out (rate when usable sites run out: 6%; usable 2026 sites: 5)

Rank	Test	Scenario	Hazard	Result
63	10	Rate at which demand needs more sites than are usable – Kauaʻi (tsunami)	Tsunami	1,265 people who may need a shelter at the rate where usable sites run out (rate when usable sites run out: 14%; usable 2026 sites: 5)
64	4	Tsunami and coastal flooding together – Kauaʻi	Tsunami + Coastal flooding	1,086 people who may need a public shelter (likely) (2026 sites usable for every hazard in the pair: 5; site gap: 0)
65	6	Wildfire and power loss together – Kauaʻi	Wildfire + Power loss	1,082 people who may need a public shelter (likely) (outlets per day if the power is out: 111)
66	11	Stays of 3 to 30 days – Kauaʻi (wildfire)	Wildfire	1,082 people who may need a public shelter (likely)
67	11	Stays of 3 to 30 days – Kauaʻi (inland flooding)	Inland flooding	1,070 people who may need a public shelter (likely)
68	11	Stays of 3 to 30 days – Kauaʻi (coastal flooding)	Coastal flooding	959 people who may need a public shelter (likely)
69	13	Kūpuna living alone, no car, far from a site – Kauaʻi (hurricane)	Hurricane	658 kūpuna living alone without a car, outside a 15-minute drive (upper bound) (Census areas outside a 15-minute drive: 7)
70	11	Stays of 3 to 30 days – Kauaʻi (tsunami)	Tsunami	452 people who may need a public shelter (likely)
71	18	Add the 2026 Point-in-Time count – Kauaʻi (coastal flooding)	Coastal flooding	440 people added by the 2026 Point-in-Time count (increase over the likely demand: 45.9%)
72	18	Add the 2026 Point-in-Time count – Kauaʻi (hurricane)	Hurricane	440 people added by the 2026 Point-in-Time count (increase over the likely demand: 20%)
73	18	Add the 2026 Point-in-Time count – Kauaʻi (inland flooding)	Inland flooding	440 people added by the 2026 Point-in-Time count (increase over the likely demand: 41.1%)
74	18	Add the 2026 Point-in-Time count – Kauaʻi (tsunami)	Tsunami	440 people added by the 2026 Point-in-Time count (increase over the likely demand: 97.3%)
75	18	Add the 2026 Point-in-Time count – Kauaʻi (wildfire)	Wildfire	440 people added by the 2026 Point-in-Time count (increase over the likely demand: 40.7%)

Rank	Test	Scenario	Hazard	Result
76	13	Kūpuna living alone, no car, far from a site – Kauaʻi (coastal flooding)	Coastal flooding	437 kūpuna living alone without a car, outside a 15-minute drive (upper bound) (Census areas outside a 15-minute drive: 3)
77	13	Kūpuna living alone, no car, far from a site – Kauaʻi (inland flooding)	Inland flooding	437 kūpuna living alone without a car, outside a 15-minute drive (upper bound) (Census areas outside a 15-minute drive: 3)
78	12	Power out 1 to 14 days – Kauaʻi (hurricane)	Hurricane	374 power-dependent residents in the risk area (outlets per day while the power is out: 374)
79	23a	No car, inside the tsunami zone, over 15 minutes' walk to a safe zone – Kauaʻi	Tsunami	292 people without a car more than 15 minutes' walk from a tsunami safe zone (Census areas over 15 minutes' walk: 1; longest walk (minutes): 16)
80	13	Kūpuna living alone, no car, far from a site – Kauaʻi (tsunami)	Tsunami	256 kūpuna living alone without a car, outside a 15-minute drive (upper bound) (Census areas outside a 15-minute drive: 5)
81	2	Lose Kalāheo Neighborhood Center and Kauaʻi High School Gym (tsunami)	Tsunami	226 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 3; people more than losing either site alone: 88)
82	14	Hearing difficulty and outside siren range (tsunami, at night) – Kauaʻi	Tsunami	199 people with hearing difficulty outside rated siren range (upper bound)
83	2	Lose Kauaʻi High School Gym and Kīlauea Neighborhood Center (tsunami)	Tsunami	183 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 3; people more than losing either site alone: 45)
84	1	Lose Kauaʻi High School Gym (tsunami)	Tsunami	138 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2)

Rank	Test	Scenario	Hazard	Result
86	2	Lose Kalāheo Neighborhood Center and Kīlauea Neighborhood Center (tsunami)	Tsunami	133 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2; people more than losing either site alone: 45)
87	12	Power out 1 to 14 days – Kauaʻi (inland flooding)	Inland flooding	114 power-dependent residents in the risk area (outlets per day while the power is out: 114)
88	12	Power out 1 to 14 days – Kauaʻi (wildfire)	Wildfire	111 power-dependent residents in the risk area (outlets per day while the power is out: 111)
89	15	Power-dependent and far from a site – Kauaʻi (hurricane)	Hurricane	107 power-dependent residents outside a 15-minute drive (Census areas outside a 15-minute drive: 7)
90	12	Power out 1 to 14 days – Kauaʻi (coastal flooding)	Coastal flooding	98 power-dependent residents in the risk area (outlets per day while the power is out: 98)
91	1	Lose Kalāheo Neighborhood Center (tsunami)	Tsunami	88 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1)
93	15	Power-dependent and far from a site – Kauaʻi (coastal flooding)	Coastal flooding	52 power-dependent residents outside a 15-minute drive (Census areas outside a 15-minute drive: 3)
94	15	Power-dependent and far from a site – Kauaʻi (inland flooding)	Inland flooding	52 power-dependent residents outside a 15-minute drive (Census areas outside a 15-minute drive: 3)
95	12	Power out 1 to 14 days – Kauaʻi (tsunami)	Tsunami	46 power-dependent residents in the risk area (outlets per day while the power is out: 46)
96	1	Lose Kīlauea Neighborhood Center (tsunami)	Tsunami	45 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1)
98	13	Kūpuna living alone, no car, far from a site – Kauaʻi (wildfire)	Wildfire	40 kūpuna living alone without a car, outside a 15-minute drive (upper bound) (Census areas outside a 15-minute drive: 1)

Rank	Test	Scenario	Hazard	Result
99	15	Power-dependent and far from a site – Kauaʻi (tsunami)	Tsunami	30 power-dependent residents outside a 15-minute drive (Census areas outside a 15-minute drive: 5)
100	15	Power-dependent and far from a site – Kauaʻi (wildfire)	Wildfire	12 power-dependent residents outside a 15-minute drive (Census areas outside a 15-minute drive: 1)
124	17	Pet households vs pet-friendly sites – Kauaʻi (hurricane)	Hurricane	488 households with a dog or a cat in shelter (usable sites that took pets in 2026 (count): 5)
125	17	Pet households vs pet-friendly sites – Kauaʻi (inland flooding)	Inland flooding	248 households with a dog or a cat in shelter (usable sites that took pets in 2026 (count): 5)
126	17	Pet households vs pet-friendly sites – Kauaʻi (wildfire)	Wildfire	244 households with a dog or a cat in shelter (usable sites that took pets in 2026 (count): 5)
127	17	Pet households vs pet-friendly sites – Kauaʻi (coastal flooding)	Coastal flooding	215 households with a dog or a cat in shelter (usable sites that took pets in 2026 (count): 5)
128	17	Pet households vs pet-friendly sites – Kauaʻi (tsunami)	Tsunami	105 households with a dog or a cat in shelter (usable sites that took pets in 2026 (count): 5)
129	19	Childcare centres in the hurricane zone – Kauaʻi	Hurricane	26 childcare centres in the zone (childcare centres in the county: 26)
130	19	Childcare centres in the wildfire zone – Kauaʻi	Wildfire	10 childcare centres in the zone (childcare centres in the county: 26)
131	19	Childcare centres in the tsunami zone – Kauaʻi	Tsunami	8 childcare centres in the zone (childcare centres in the county: 26)
132	19	Childcare centres in the coastal flooding zone – Kauaʻi	Coastal flooding	6 childcare centres in the zone (childcare centres in the county: 26)
133	19	Childcare centres in the inland flooding zone – Kauaʻi	Inland flooding	6 childcare centres in the zone (childcare centres in the county: 26)
134	16	Chuukese: 0 of 15 messages translated – Kauaʻi	All hazards	people who speak English less than very well: needs county data (messages translated: 0)
135	16	Marshallese: 0 of 15 messages translated – Kauaʻi	All hazards	people who speak English less than very well: needs county data (messages translated: 0)
136	22	Care facilities in the coastal flooding zone – Kauaʻi	Coastal flooding	0 care facilities in the zone (facilities over 15 minutes from a usable site, or none on the island: 0)

Rank	Test	Scenario	Hazard	Result
137	22	Care facilities in the inland flooding zone – Kauaʻi	Inland flooding	0 care facilities in the zone (facilities over 15 minutes from a usable site, or none on the island: 0)
138	22	Care facilities in the tsunami zone – Kauaʻi	Tsunami	0 care facilities in the zone (facilities over 15 minutes from a usable site, or none on the island: 0)

Questions for counties

These need county knowledge. The plan has no data for them, so it shows no numbers.

1. **Shelter buildings in an earthquake or a strong hurricane:** Which shelter buildings have been assessed for earthquake and hurricane-wind resilience, and which need a check before they are used after a strong event?
2. **Resupply for Molokaʻi and Lānaʻi:** How would water, food, cots, fuel and medical supplies reach Molokaʻi and Lānaʻi if the harbours or airports close for several days, and who decides the order of shipments?
3. **Cell network outages:** Which areas lose mobile phone service first in a long power outage or after wind damage, and how will warnings and shelter information reach them (radio, sirens, door-to-door)?
4. **Schools used as shelters:** When a school is used as a shelter, how long can it stay open before classes must restart, and where do residents go next?
5. **Partner capacity:** Could each county collect partner capacity (cots, meals, staff, vehicles, interpreters) with a county-run form such as Survey123, with no personal data, so the plan can show pledged capacity instead of 'confirm with partner'?